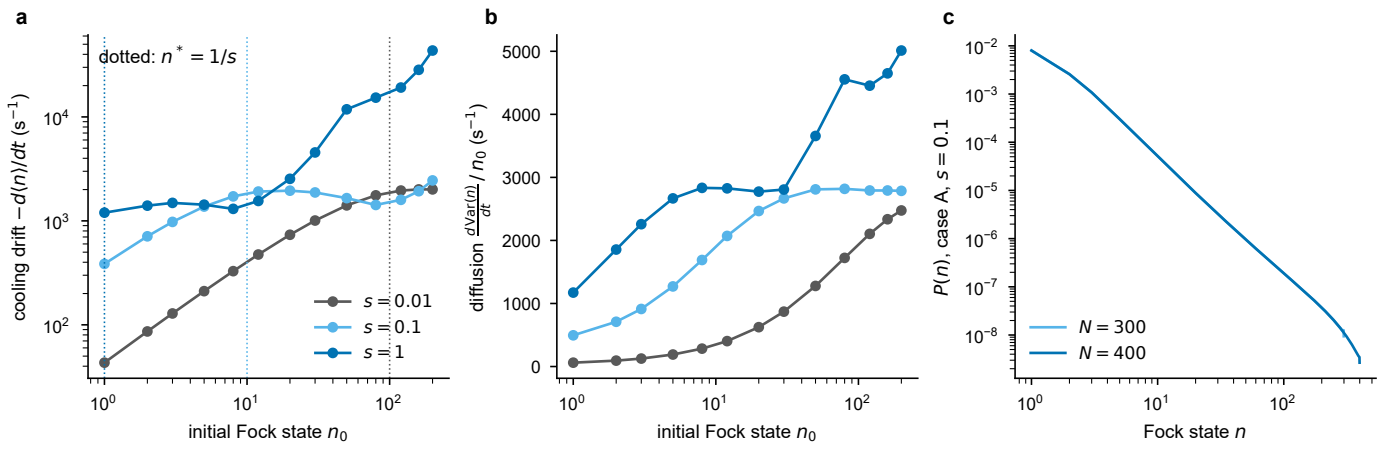

a
dotted: $n^* = 1/s$
cooling drift $-d\langle n\rangle/dt$ (s^{-1})
10^4
10^3
10^2
$s = 0.01$
$s = 0.1$
$s = 1$
10^0
10^1
10^2
initial Fock state n_0
b
diffusion $\frac{d\mathrm{Var}\langle n\rangle}{dt} / n_0$ (s^{-1})
5000
4000
3000
2000
1000
0
10^0
10^1
10^2
initial Fock state n_0
c
$P(n)$, case A, $s = 0.1$
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
$N = 300$
$N = 400$
10^0
10^1
10^2
Fock state n